

Module 4, Assignment 3: Output Check

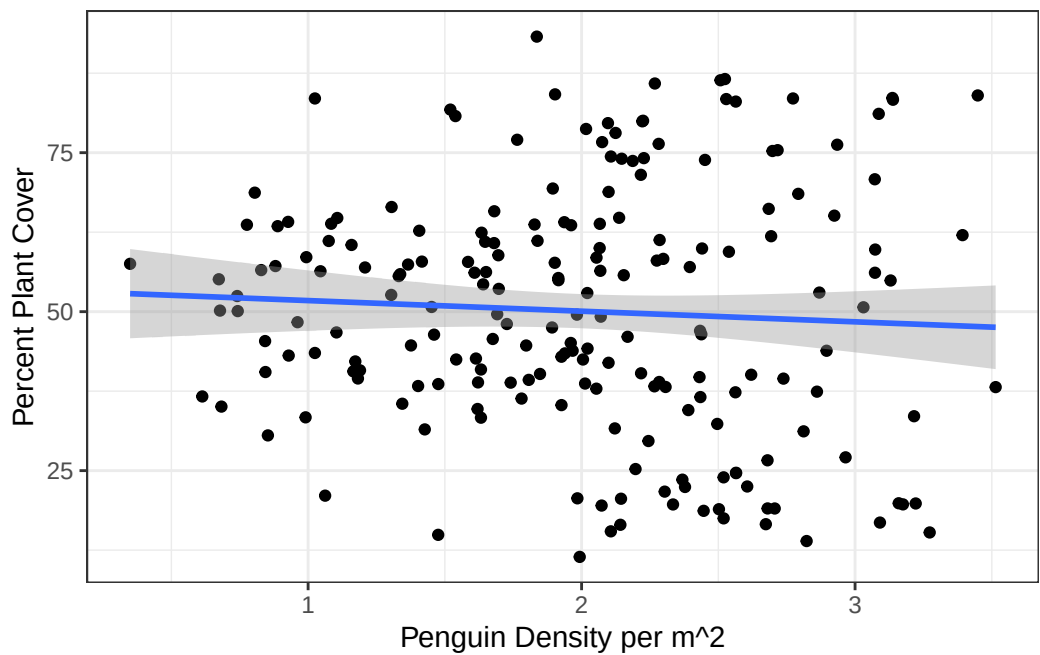
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3.

```
# A tibble: 6 x 4
  site plant_type percent_plant_cover penguin_density_m2
<dbl> <chr>          <dbl>          <dbl>
1     1 moss          47.5            1.89
2     2 moss          39.5            1.18
3     3 moss          39.3            1.81
4     4 moss          40.9            1.63
5     5 moss          45.4            0.843
6     6 moss          36.7            0.613
```

```
# A tibble: 6 x 4
  site plant_type percent_plant_cover penguin_density_m2
<dbl> <chr>          <dbl>          <dbl>
1   195 liverwort    22.4            2.38
2   196 liverwort    24.6            2.56
3   197 liverwort    19.1            2.68
4   198 liverwort    31.6            2.12
5   199 liverwort    20.6            1.98
6   200 liverwort    11.4            1.99
```

5.



6.

```
[1] -0.05922145
```

7.

```
[1] 0.00350718
```

8.

Call:

```
lm(formula = percent_plant_cover ~ penguin_density_m2, data = plants)
```

Residuals:

Min	1Q	Median	3Q	Max
-38.647	-12.154	-0.661	12.233	42.906

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	53.413	4.212	12.682	<2e-16 ***
penguin_density_m2	-1.670	2.001	-0.835	0.405

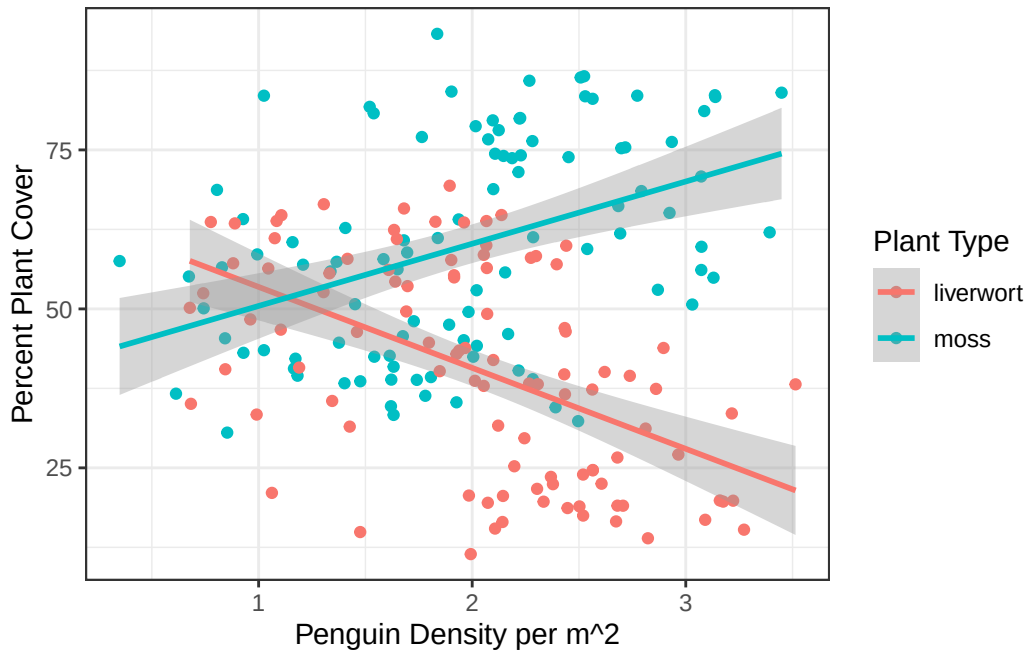
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 19.29 on 198 degrees of freedom

Multiple R-squared: 0.003507, Adjusted R-squared: -0.001526

F-statistic: 0.6969 on 1 and 198 DF, p-value: 0.4048

9.



10.

Call:

```
lm(formula = percent_plant_cover ~ penguin_density_m2 * plant_type,  
    data = plants)
```

Residuals:

Min	1Q	Median	3Q	Max
-32.757	-12.508	1.299	11.637	34.609

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	66.183	4.776	13.858	< 2e-16 ***

penguin_density_m2	-12.730	2.238	-5.688	4.62e-08	***
plant_typemoss	-25.508	6.508	-3.920	0.000123	***
penguin_density_m2:plant_typemoss	22.516	3.089	7.289	7.51e-12	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 14.86 on 196 degrees of freedom

Multiple R-squared: 0.4151, Adjusted R-squared: 0.4061

F-statistic: 46.37 on 3 and 196 DF, p-value: < 2.2e-16